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Paleomagnetically defined rotations from the Precordillera of northern Chile: Evidence of localized in situ fault-controlled rotations

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Abstract. A paleomagnetic study of Paleocene, Cretaceous, and Jurassic units in the Precordillera of northern Chile (26°S to 27°S) demonstrate a complex pattern of crustal rotations. The region was selected to investigate the pattern of deformation associated with the Sierra Castillo-Agua Amarga Fault and its associated structural subdomains, which form part of the Domeyko Fault System, the fault system that controls much of the structure of the Chilean Precordillera. Paleocene lavas from the center of the study area pass a fold test, indicating the primary nature of the remanence in these rocks. A second group of lavas from the south has a similar formation mean inclination after tilt correction for a uniform dip, which coupled with the presence of reversals, suggests that these lavas also carry a primary remanence. The data indicate clockwise rotations of ~35° and 42° respectively, for the two sampling localities. Three geographically separate areas of Lower Cretaceous red beds also reveal primary remanences based on the presence of sites with reversed polarity at two of the localities and an inclination-only fold test of the three locality mean directions for this formation. These sampling areas yield remanence directions that reveal 30° of counterclockwise and 31° of clockwise rotation as well as an area without statistically significant rotation. These data include the first well-constrained counterclockwise rotation to be recovered from rocks in the southern central Andes, although mapping suggests the block involved is of limited geographic extent. Jurassic strata failed to yield any reliable results. Overall, the data indicate a greater variability in the rotation pattern than has previously been observed in most comparable areas south of the Arica Deflection. The variation cannot be explained by large-scale rotation mechanisms, such as oroclinal bending or domino-style block rotation by widely separated transandean faults. Instead, the rotations are consistent spatially, and temporally and in their sense and magnitude with the Eocene transpressional deformation associated with the Domeyko Fault System; the deformational event that generated most of the important structures in the studied area. The data emphasize the significance of local tectonics in controlling rotations in this part of the south central Andes and the importance of understanding the local structure of regions sampled for paleomagnetic study.

1. Introduction

Paleomagnetic investigations carried out in the central Andes have identified a fairly consistent pattern of clockwise rotations to the south of the Arica Deflection (Figure 1a) and counterclockwise rotations to the north [e.g., *Beck*, 1988, 1998; *Roperch and Carlier*, 1992; *Somoza et al.*, 1996; *Randall*, 1998]. Individual rotations and the large-scale pattern observed have been attributed to various tectonic models and mechanisms. At the center of controversy in the differing models is the role of large-scale versus local processes in the generation of the overall rotation pattern. Large-scale mechanisms include forearc wide rotations due to oroclinal bending produced by differential shortening [*Isacks*, 1988; *Roperch and Carlier*, 1992; *Butler et al.*, 1995; *MacFadden et al.*, 1995] and large-scale domino-style block rotation by widely separated transandean basement faults [*Abels and Bischoff*, 1999]. Local processes include rotations of smaller-scale crustal blocks in response to oblique subduction [*Beck*, 1988; *Hartley et al.*, 1988; *Forsythe and Chisholm*, 1994; *Randall et al*, 1996; *Somoza et al*, 1996; *Taylor et al*, 1998] and rotation of detached or partially detached thrust sheets [*Hartley et al.*, 1992a, *Butler et al.*, 1995].

This paper presents results from a paleomagnetic study in the Andean Precordillera of northern Chile. The study area lies between 26° and 27°S and 69°15' and 69°40'W, in the region of the major mining districts of El Salvador and Potrerillos (Figure 1). Jurassic to Paleocene volcanic rocks and Cretaceous sedimentary units were sampled for

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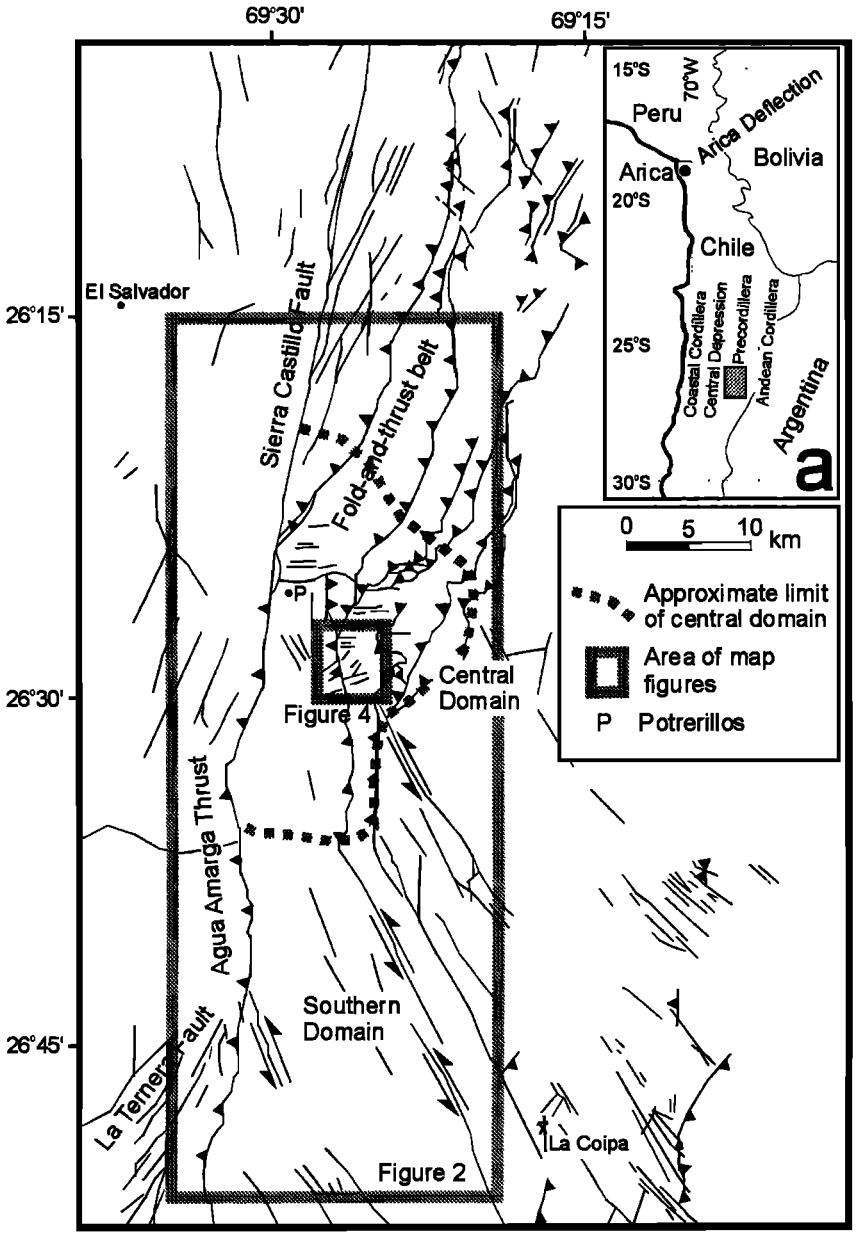


Figure 1. Map of major faults in the study area. Inset a shows position of study area within northern Chile.

paleomagnetic study in order to detect any rotations, constrain the timing of these rotations, and if possible, relate the rotations to the known structural evolution of the area. The investigation was designed to assess the magnitude and distribution of rotations within an area that has faults with well-constrained kinematics and movement ages [*Cornejo et al.*, 1993, 1997; *Tomlinson et al.*, 1993; Niemeyer, 1999]. Combining paleomagnetic and structural data allows the relationships between deformation and block rotation to be identified and in the current study indicates that localized in situ fault-controlled rotations dominate the pattern of rotations in the Precordillera of northern Chile between 26°S and 27°S.

2. Regional Geology

The geology of this part of the Precordillera is the consequence of four distinct tectonomagmatic episodes in the evolution of the Chilean margin which young successively and episodically eastward. During the Late Triassic to Early Cretaceous the magmatic arc lay in th Coastal Cordillera, and the Precordillera formed part of an extensive back arc to intra-arc basin system [*Mpodozis and Ramos*, 1989; *Scheuber and Reutter*, 1992; *Ardill et al.*, 1998]. During the mid-Cretaceous the magmatic arc migrated eastward to occupy the present-day Central Depression [*Scheuber and Reutter*, 1992;

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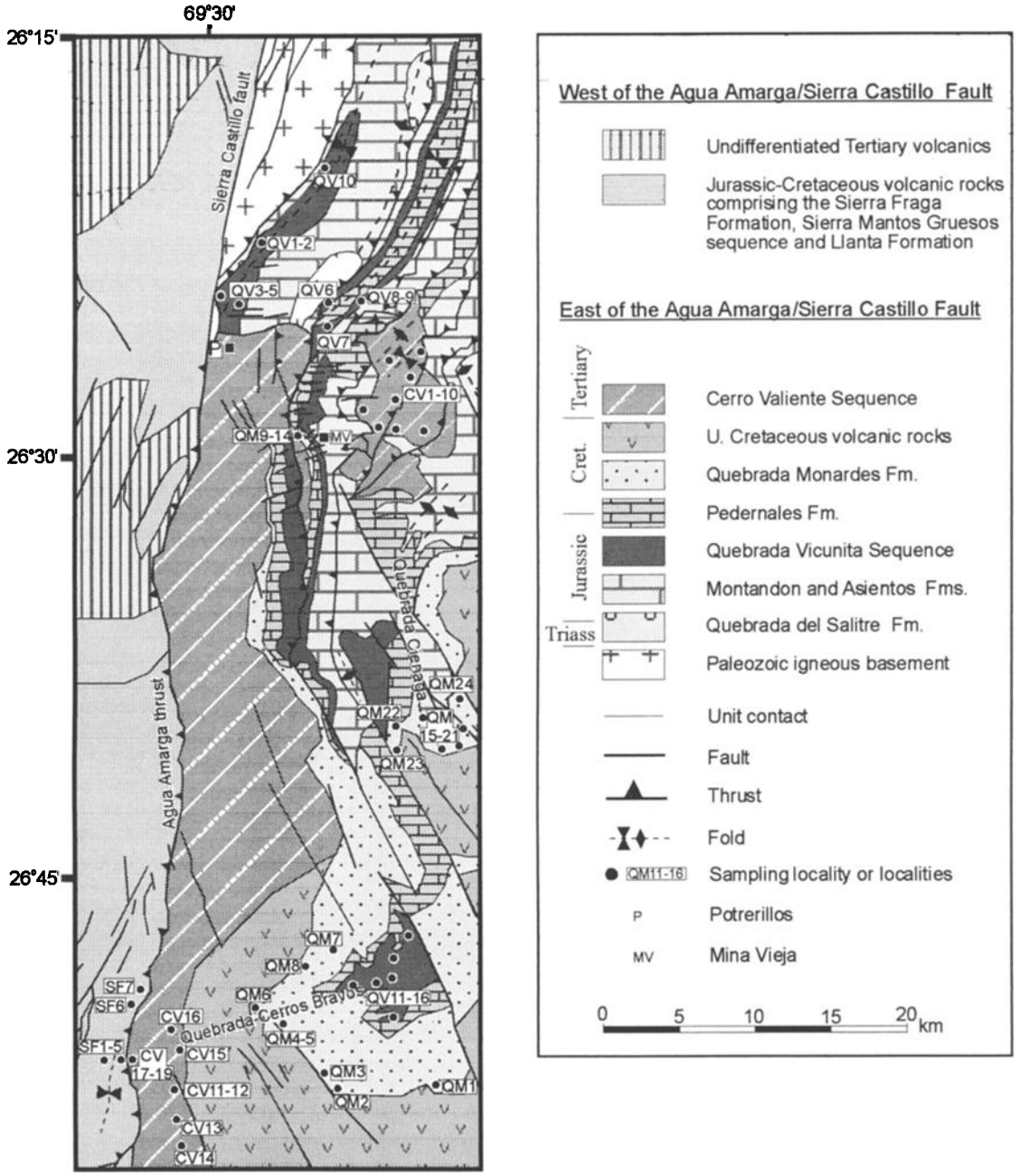


Figure 2. Simplified geological map of the study area showing sampling localities (redrawn from *Cornejo et al.* [1997, 1998] and *Tomlinson et al.* [1999]).

Randall et al., 1996]. Subsequently, in the Late Cretaceous and Paleogene the Precordillera was itself the location of arc magmatism [*Cornejo et al.*, 1997], and during the Oligocene the locus of magmatism again migrated eastward to its present position in the Andean Cordillera, such that the Precordillera forms part of the present-day forearc.

The major structural feature of the studied area is the north-south trending fault system comprising the Sierra Castillo and Agua Amarga Faults (SC-AA) (Figure 1), which together with the La Ternera Fault System, to the south, and the Sierra de Varas and West Fissure Fault Systems, to the north, comprise the Domeyko Fault System [*Maksaev and Zentilli*, 1988; *Tomlinson et al.*, 1993; *Mpodozis et al.*, 1993]. The SC-AA Fault separates two areas with contrasting Mesozoic histories, indicating activity of the SC-AA Fault since the Jurassic [*Cornejo et al.*, 1993; 1997].

The Mesozoic geology west of the SC-AA Fault comprises a Jurassic-Cretaceous succession of predominantly volcanic rocks intercalated with marine limestones, terrigenous sandstones, volcaniclastic sandstones, and breccias (Figure 2).

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The succession is dominated by the ~3000 m thick Jurassic Sierra Fraga Formation [*Sepúlveda and Naranjo*, 1982], which forms the oldest exposed unit west of the SC-AA Fault. The succession has limited areal extent and contrasts in facies with coeval units east of the SC-AA Fault, suggesting deposition in an extensional basin where the SC-AA Fault was one of the major extensional faults bounding the east side of the basin [*Cornejo et al.*, 1993, 1997; *Tomlinson et al.*, 1999].

East of the SC-AA Fault the pre-middle Cretaceous Mesozoic geology is characterized by an Upper Triassic to Lower Cretaceous succession of marine and continental facies sedimentary rocks and sparse volcanic rocks in unconformable contact with Paleozoic basement comprising granitic and volcanic rocks (Figure 2) [*Cornejo et al.*, 1997, 1998; *Tomlinson et al.*, 1999]. The basal sedimentary units were deposited during back arc conditions prevailing during the early Mesozoic. The Upper Triassic to Middle Jurassic rocks comprise limestone, shale, and terrigenous and volcaniclastic sandstones of the Quebrada del Salitre, Montandón, and Asientos Formations that were deposited in a west facing platform environment [*Harrington*, 1961;

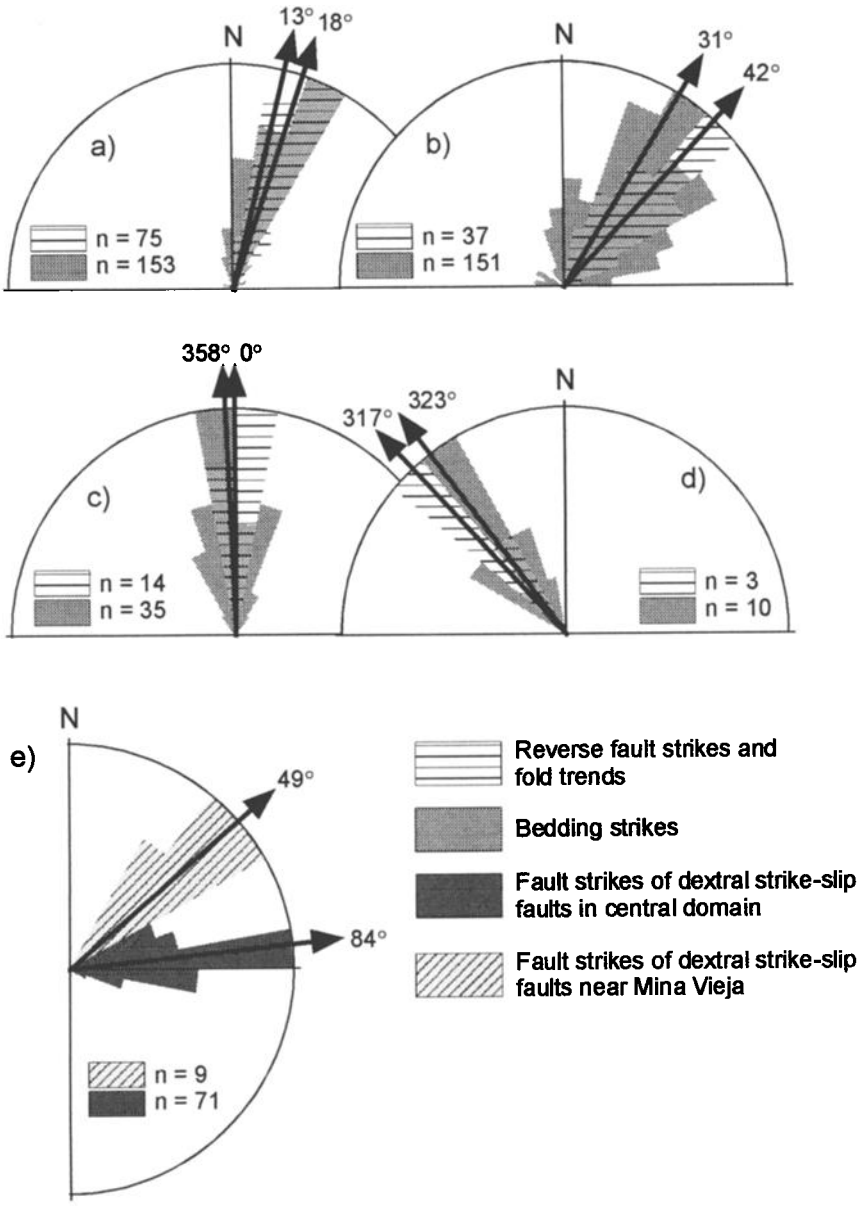


Figure 3. Rose diagrams for the strikes of bedding and faults and trends of folds in the northern and central domains. The arrows indicate the mean direction for the corresponding data. (a) northern domain, (b) central domain except its southwest sector, (c) southwest sector of the central domain excluding the area of the Mina Vieja sampling locality, (d) area of the Mina Vieja sampling locality, (e) Comparison of fault strikes for dextral strike-slip faults throughout the central domain with those at the Mina Vieja locality. In Figure 3e and comparing Figure 3d with 3c structures in the Mina Vieja area are revealed to be systematically oriented 35°-40° counterclockwise with respect to the equivalent structures in the surrounding area.

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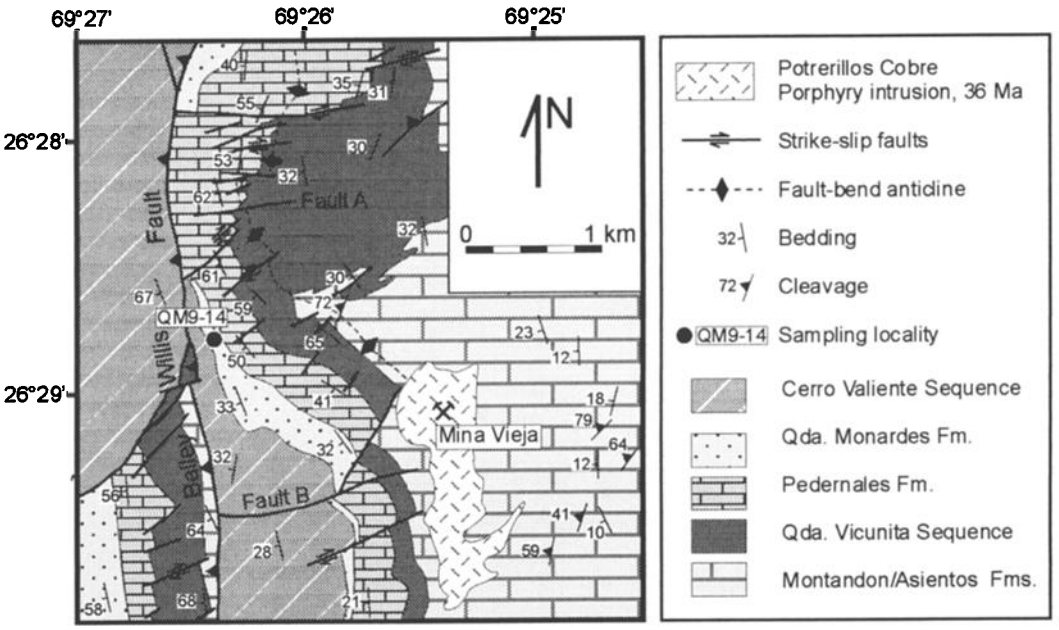


Figure 4. Geological map of the Mina Vieja area. Units and symbols are the same as those in Figure 2 except for those indicated in the legend. Fault A, fault B, and the Bailey Willis Fault, on the north, south and west, respectively, bound an area which contains bedding, fold, and fault orientations that are counterclockwise with respect to those in the surrounding area. The younger-on-older relationship across the northern part of the Bailey Willis Fault is the result of incomplete Eocene inversion of a Paleocene normal fault (A. Tomlinson, unpublished data, 1994). (Redrawn from *Tomlinson et al.* [1999]).

Cornejo et al., 1993]. These are superseded by the Upper Jurassic Quebrada Vicuñita sequence [*Cornejo et al.*, 1993, 1998], a series of basaltic-andesitic lavas intercalated with thin marine sedimentary horizons, and the Upper Jurassic-Lower Cretaceous Pedernales Formation, comprising marine limestone and volcaniclastic sandstones [*Harrington*, 1961]. Subsequently, a regional marine regression [*Ardill et al.*, 1998] led to the accumulation of fluvial and aeolian facies arkosic sandstone and coastal sabkha facies evaporites, which together form the Quebrada Monardes Formation [*Suárez and Bell*, 1985; *Bell*, 1991] and its correlative, the Agua Helada Formation [*Garcia*, 1967].

The Late Cretaceous and Paleogene magmatic arc is represented by three major phases of activity, occurring in the Late Cretaceous, Paleocene, and Eocene-Oligocene [*Cornejo et al.*, 1997]. One of the most widely exposed units from this arc is the Paleocene (63-55 Ma) Cerro Valiente sequence, a highly variable sequence of flows with lesser amounts of ignimbrites and breccias [*Cornejo et al.*, 1993, 1998]. This unit lies unconformably to paraconformably on the Lower Cretaceous sedimentary rocks and Upper Cretaceous volcanic rocks. The Eocene-Oligocene saw the emplacement of numerous, widely distributed intrusions, including shallow-level bodies associated with the economically important porphyry copper deposits. The Eocene intrusions are coeval with the last major deformational event in the Precordillera, the Incaic orogenic phase, which generated most of the important structures in the studied area [*Tomlinson et al.*, 1993; *Mpodozis et al.*, 1994; *Cornejo et al.*, 1997]. Since abandonment of this arc and subsequent eastward relocation of magmatism in the Oligo-Miocene, the Precordillera region has undergone a period of uplift.

3. Structural Geology

Although having an earlier history as a Mesozoic extensional fault, the main phase of motion on the SC-AA Fault was during the Eocene contractional deformation of the magmatic arc [*Tomlinson et al.*, 1993; *Cornejo et al.*, 1997]. At this time, the Sierra Castillo fault was reactivated as a near-vertical sinistral strike-slip fault with a component of up on the east slip [*Tomlinson et al.*, 1993; Niemeyer, 1999]. In contrast, the southern fault segment, the Agua Amarga thrust, is a westside up oblique reverse fault. Despite the opposite senses of vertical movements on these two fault segments, *Tomlinson et al.* [1993] argued that they are contemporary and form a single master fault for a sinistral transpressive system kinematically linked to the broad zone of deformation east of the SC-AA Fault.

To the east of the SC-AA Fault are three distinct but kinematically related structural domains of subsidiary faults (Figure 1) [*Tomlinson et al.*, 1993]. (1) The Northern domain consists of a fold-and-thrust belt which can be subdivided into two subdomains based on structural style: an eastern subdomain characterized by thin-skinned, ramp flat style folding and thrusting and a western subdomain displaying thick-skinned, fault propagation fold style deformation involving the Paleozoic basement (Figure 2). Bedding,

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reverse faults, and folds in the domain have strikes and trends of ~10-25° (Figure 3a). (2)The Central domain is located at the southernmost end of the fold-and-thrust belt and is a complex region of thrusting and folding associated with numerous dextral E-W strike-slip faults. In this domain, the lower, easterly positioned thrust faults have decreasing displacements along strike to the southwest and die into tip line folds or terminate in the eastern ends of dextral E-W strike-slip faults which transfer the thrust displacement to the next higher and westerly positioned thrust. Other thrust faults unite directly with higher thrusts. As a consequence, the shortening on lower thrusts is progressively replaced to the southwest by added movement on higher thrusts. In this manner, the domain is interpreted to be a displacement transfer zone in which shortening across a broad zone in the northern domain is transferred to a few higher thrusts, particularly the Agua Amarga thrust. Over much of the central domain the regional structural features have orientations of 030°-045° (Figure 3b). In the southwest sector of the domain, where the higher reverse faults dominate the shortening, the structural trends shift to a N-S orientation (Figure 3c), except in a ~2 x 2 km area near Mina Vieja where the structures trend 35°-40° counterclockwise with respect to those in the surrounding area (Figures 3d, 3e, and 4). This observation led *Cornejo et al.* [1993] to predict that this small block had experienced counterclockwise rotations. (3) The Southern domain is characterized by sinistral strike-slip faults with a NW orientation. The NW trending faults with greatest continuity and displacement occur in the Quebrada Ciénaga area (Figure 2) and connect the southern end of reverse faults in the central domain to NNE oriented high-angle reverse faults in the La Coipa area (Figure 1). They are interpreted to represent another transfer zone [*Cornejo et al.*, 1993]. Outside this transfer zone the faulting is minor, and instead, deformation is dominated by shortening related to the Agua Amarga thrust.

Syntectonic and post-tectonic intrusions associated with the faults indicate that regional deformation was initiated by 42 Ma, continued until at least 36 Ma, and ended by 32 Ma [*Tomlinson et al.*, 1993; *Mpodozis et al.*, 1994; *Cornejo et al.*, 1997]. The SC-AA Fault and associated domains define a wide deformation zone with structure orientations and kinematics consistent with development in response to sinistral transpression where the SC-AA acted as the master fault to the structural system. In most types of transpression, clockwise and counterclockwise rotation of passive markers of different orientations are expected [*Fossen and Tikoff*, 1998]. Folds formed at an angle to the master fault rotate toward it with progressive transpression [*Sanderson and Marchini*, 1984; *Jamison*, 1991]; therefore counterclockwise rotations may be anticipated in the fold-and-thrust belt of the northern domain. In the southern domain, if slip on the NW faults accommodates rotation, these faults and the intervening blocks might rotate either clockwise or counterclockwise depending on the relative importance of the pure and simple shearing components in the deformation history, with system-perpendicular pure shearing (contraction) producing sinistral slip on the NW faults and clockwise rotations and system-parallel sinistral simple shearing generating dextral slip on the NW faults and counterclockwise rotations. Considering the evident importance of shortening in the system and the sinistral slip sense of the NW faults, a clockwise rotation is expected. The central domain is more complex in that it is part of the fold-and-thrust belt and so could have counterclockwise rotations but also has a component of dextral E-W simple shearing related to displacement transfer in the locus of shortening and so could have clockwise rotations.

4. Paleomagnetic Sampling and Analytical Procedures

Samples from 66 sites were collected and analyzed from the central and southern structural domains east of the SC-AA Fault and from a small area above the Agua Amarga thrust, west of the main trace of the SC-AA Fault (Figure 2). Samples were collected from the Cerro Valiente (Paleocene) and Quebrada Vicuñita (Upper Jurassic) sequences and from the Quebrada Monardes (Lower Cretaceous) and Sierra Fraga (Jurassic) Formations. The Cerro Valiente and Quebrada Vicuñita sequences were sampled from the southern and central domains. The Quebrada Monardes Formation was sampled in the southern domain and at Mina Vieja in the central domain, where the structures trend counterclockwise with respect to the equivalent structures in the surrounding area. The northern domain was not sampled because of its inaccessibility, and only the Sierra Fraga Formation could be sampled from west of the SC-AA Fault (Figure 2).

Natural remanent magnetizations (NRM) were measured mainly using a Molspin MS2 or Geofyzika JR4 spinner magnetometer. A CCL cryogenic magnetometer was also used for some of the Quebrada Monardes Formation samples. Specimens were stepwise demagnetized by thermal or two-axis tumbling alternating field (AF) methods through at least 12 steps until they became unstable, or to 700°C or 100 mT, respectively. During thermal demagnetization, bulk susceptibility was measured after each thermal demagnetization step to detect heat-induced mineral changes. Remanence directions were determined using principal component analysis [*Kirschvink*, 1980]. Components were defined by a minimum of three points on vector endpoint diagrams and were considered stable when they had a maximum angular deviation (MAD) not exceeding 15°, although in most cases the observed MAD were < 10°. In some cases, multiple magnetic components with overlapping blocking or coercivity spectra required the use of remagnetization circles combined with endpoints [*McFadden and McElhinny*, 1988] to isolate the characteristic remanent magnetization (ChRM). Mean directions for each geological unit and sampling locality were calculated by combining site mean directions [*Fisher*, 1953].

5. Paleomagnetic Results

5.1 Cerro Valiente Sequence (Paleocene)

A total of 19 sites, mostly oriented hand samples, were collected and demagnetized from the Cerro Valiente sequence, 10 (CV1-10) from the central and 9 (CV11-19)

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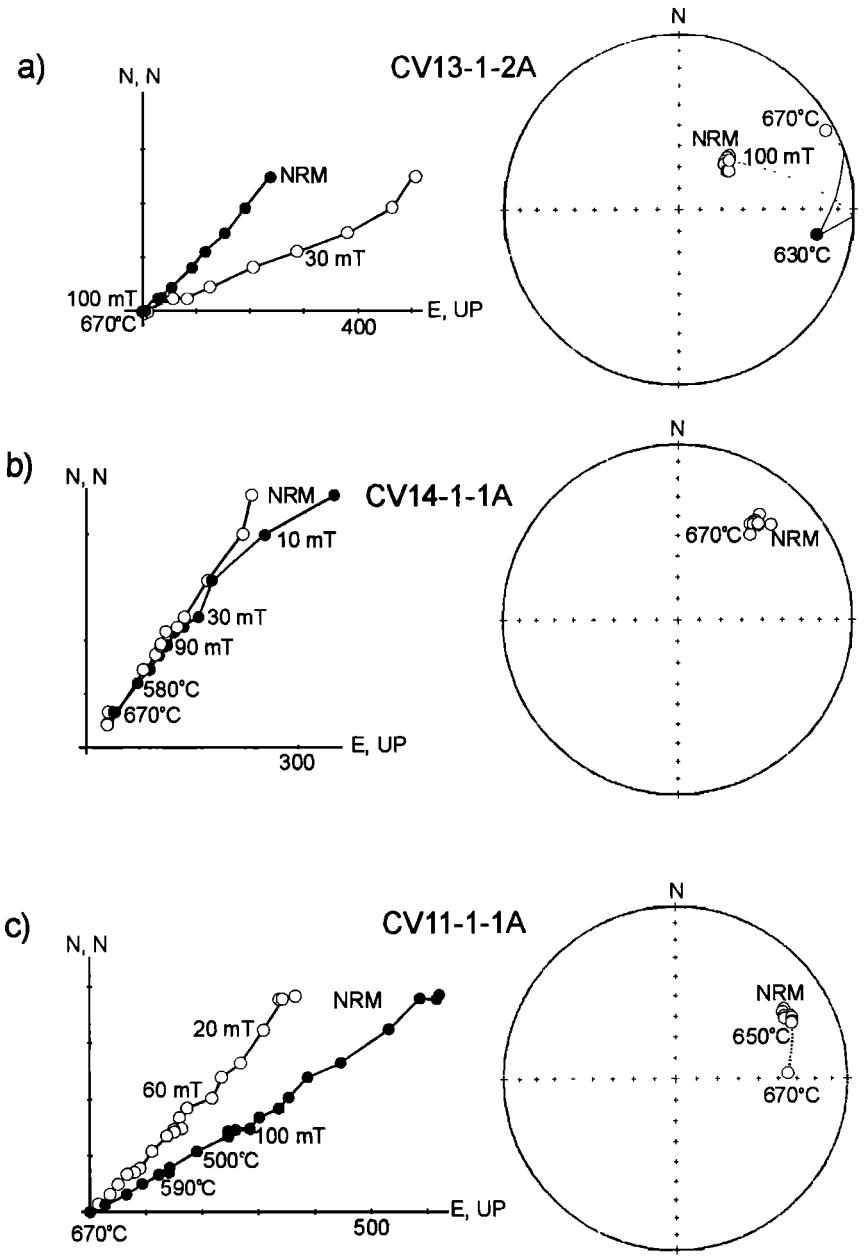


Figure 5. Representative orthogonal and equal-area projections for samples from the Cerro Valiente sequence. In orthogonal plots, open (solid) circles represent vertical (horizontal) projections; intensities are in mAm^{-1}. On stereoplots, open (solid) circles represent negative (positive) inclinations. All plots are shown tilt corrected.

from the southern structural domains (Figure 2). Where hand samples were taken, at least two independently oriented samples were collected, from which two or three oriented cores with different orientations were drilled. In the central domain, sites were collected from around an open, noncylindrical, NE-SW oriented syncline, while in the southern domain, sites were collected from a sequence of lavas of fairly consistent dip. Paleohorizontal control was obtained from igneous flow surfaces.

NRM intensities are in the range 250-2400 mAm^{-1}, although most samples fall in a 400-800 mAm^{-1} range and are broadly consistent within individual sites. All sites display simple demagnetization behavior, whereby a single component of magnetization is readily isolated after removal of a low-coercivity component by treatment to ~10-60 mT (Figure 5). The low-coercivity components are scattered both within and between sites and are interpreted as representing short-term laboratory-acquired viscous remanent magnetizations (VRM). Demagnetization of the NRM suggests that high- and low-coercivity magnetic mineral phases are present in the samples, interpreted as magnetite and hematite. AF demagnetization reduced the NRM to

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Table 1. Results From the Cerro Valiente Sequence[a]

		Uncorrected			Tilt Corrected			
Site	n/N	Dec.	Inc.	TC	Dec.	Inc.	k	α95
				Central Domain				
CV1[b]	5/6	232.9	64.4	210/43	271.7	33.0	18.6	19.5
CV2	5/5	206 0	48.9	121/35	241.9	41.0	1560.0	1.9
CV3	3/6	196.7	17.3	260/30	206.0	43.2	47.4	18.1
CV4[b]	6/6	194 5	51.6	320/35	134.4	68 8	72.1	8.5
CV5	3/3	243.3	27.0	355/29	231.8	53.0	97.4	14 7
CV6[b]	6/6	222.5	20 2	208/61	232.6	-2.2	51.8	9.4
CV7	6/6	303.9	55.0	090/77	213 2	29.7	72.6	8 5
CV8	5/7	8.4	6.0	251/46	14.6	-34.2	52.4	10.6
CV9	4/7	221.7	6.1	259/46	235.1	30.9	73 4	10 8
CV10	7/8	36.3	-28.4	330/8	40.7	-30.0	14.3	16.5
Mean IS	7/10	38.1	-28.7				5.1	29.5
Mean TC					40.1	-38.5	26.0	12.1
				Southern Domain				
CV11	6/6	24.0	-57 7	190/33	57.6	-39.7	26.6	13.2
CV12	6/6	19.9	-45.5	202/31	48.2	-38.7	56.3	9.0
CV13	6/6	356.3	-51.3	198/36	43.1	-50.1	54.4	9 2
CV14	6/6	16.3	-38.4	196/36	41.2	-30.0	207.5	4.7
CV15	6/6	357.0	-77.6	180/27	63.7	-61.1	271.4	4.1
CV16	6/6	36.6	-52.3	226/10	49.7	-52.8	352 7	3.6
CV17	7/7	177.9	71.5	180/50	246.2	38.2	27.2	11.8
CV18	6/6	10.8	-49.3	180/50	45.9	-23.2	41.6	11.3
CV19	6/7	173.2	32.0	180/50	202.1	24.7	156.7	5.4
Mean IS	9/9	11.2	-53.6				22.7	11.0
Mean TC					47.5	-40.5	24.2	10.7

[a] Notation is as follows: n, the number of samples contributing to the mean; N, the total number of demagnetized samples for the site; Dec. and Inc., the declination and inclination of the site mean magnetization vector (in degrees), TC, the tilt correction, given as strike and dip; k and α95, the statistical parameters of *Fisher* [1953]. Mean IS is the in situ mean direction and Mean TC is the tilt-corrected mean direction.
[b] Indicates sites excluded from the site mean calculation.

between 40 and 10% of its initial intensity and was the preferred treatment method as many samples broke up when heated. When a direction could be recovered from the hematite component, it was found to have the same direction as the component carried by magnetite.

Of the 10 sites from the central domain, seven carry a consistent tilt-corrected direction, taken to be the ChRM, directed to the NE with moderate inclination or its antipode (Table 1 and Figure 6). Three sites excluded from the formation mean all have remanence directions that lie > 2σ away from the ChRM after tilt correction. The grouping of the seven sites is significantly better after tilt correction and passes a fold test at the 99% confidence level [*McElhinny*, 1964]. All nine sites from the southern domain carry a ChRM, again directed to the NE with moderate upward inclination or, in two sites, its antipode (Table 1 and Figure 6). The uniformity of the bedding corrections means that a meaningful fold test cannot be performed on this data. Application of a reversal test [*McFadden and McElhinny*, 1990] yields an indeterminate result for both study areas at the 95% confidence level, probably a consequence of the uneven distribution of normal and reverse polarity sites at each sampling locality. In both cases the angle between the polarity groups is smaller in tilt-corrected than in situ coordinates.

5.2 Quebrada Monardes Formation (Lower Cretaceous)

Twenty-four sites were collected from the Quebrada Monardes Formation (Figure 2) as oriented hand samples and field-drilled cores. Sites QM1 to QM8 were collected from a gently SW plunging, open anticline in the south of the study area near Quebrada Cerros Bravos. Sites QM15 to QM24 were collected farther north, from a NW trending anticline in Quebrada Ciénaga. Both of these sampling localities lie within the southern domain of NW trending strike-slip faults. Sites QM9 to QM14 come from a 20 m plus thick section with approximately constant dip near Mina Vieja (Figure 4) in an area having anomalous structural trends in the southwest sector of the central domain, as previously described.

The total NRM intensities for these red bed samples range between 0.5 and 187 mAm^{-1}, with the majority < 50 mAm^{-1}. Demagnetization behavior in the samples is variable and typically multicomponent (Figures 7a - 7d). In some cases the components have overlapping blocking spectra (Figure 7d), and the ChRM is only recovered as a linear segment in two or three samples from the site, while others only define remagnetization circles. When clearly isolated, low unblocking temperature components tend to be either present-day field directions, interpreted as long-term VRM, or are dispersed, representing laboratory-induced VRM. Despite the

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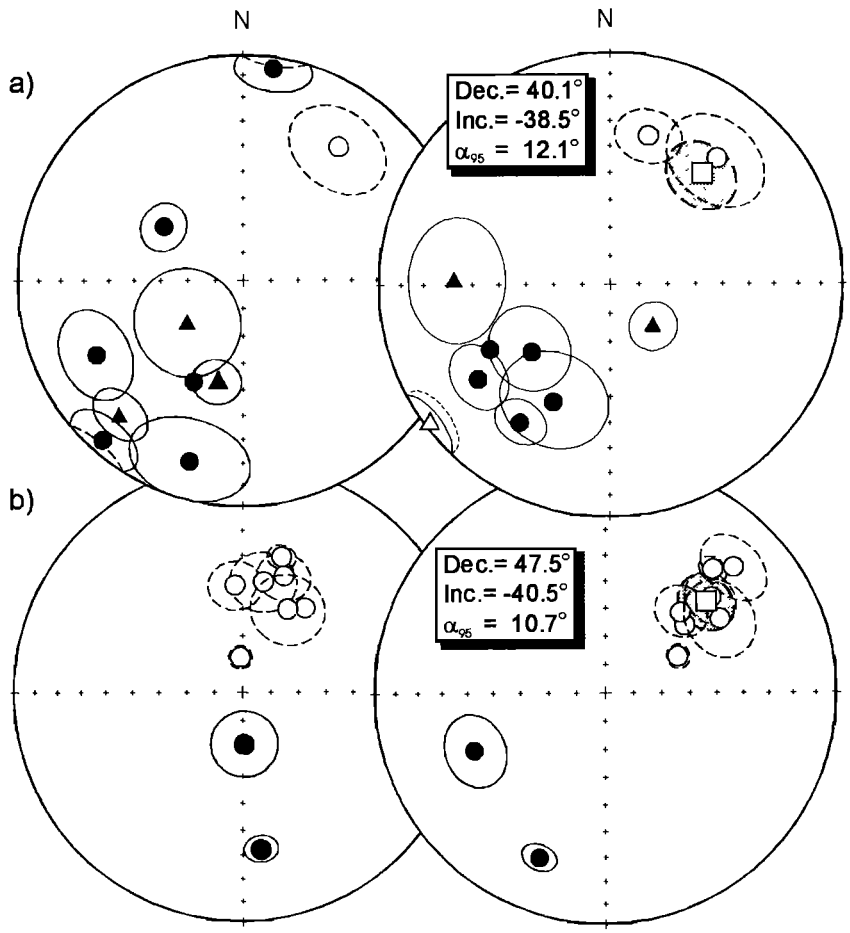


Figure 6. Equal-area projections summarizing the paleomagnetic data from the Cerro Valiente sequence. The data are shown in (left) in situ (geographic) coordinates and (right) tilt-corrected coordinates. (a) central domain and (b) southern domain Open (solid) circles represent negative (positive) inclinations with their associated 95% confidence limits. The formation mean direction is represented by a square, and its 95% confidence limit is shaded. Triangles denote sites excluded from the formation mean.

complicated nature of the NRM, a consistent, well-defined ChRM was recovered from all sites.

Sites from the Quebrada Cerros Bravos region yielded a consistent ChRM, directed upward in the NE quadrant or, in two sites, its antipode (Table 2 and Figure 8a). Where stable endpoints were recovered, it was through thermal treatment above 580°C. All but one of the 10 sites analyzed from the Quebrada Ciénaga locality carried both normal and reversed polarity components (Table 3 and Figure 8b), with the reversed polarity component only isolated by demagnetization above 580°C (Figure 7d). The mean directions of the two polarity groups are antipodal at the 95% confidence level, both in situ and tilt corrected, passing the reversal test with a C classification in both cases [*McFadden and McElhinny*, 1990]. Thermal demagnetization to temperatures > 580°C isolated a consistent normal polarity remanence direction, directed upward in the NW quadrant from all six sites collected from near Mina Vieja (Table 2 and Figure 8c). One site (QM12) has a relatively poorly defined site mean direction ($\alpha_{95} > 20°$), but it is broadly consistent with the other sites and hence is included in the overall group mean.

The positive reversal test demonstrates that the directional difference between the two polarities groups at the Quebrada Ciénaga locality is small, so either group mean could be used for the tectonic analysis. However, the direction recovered from the hematite component has an inclination closer to that expected at this sampling latitude. Furthermore, the remanence recovered from the Quebrada Monardes Formation at Quebrada Cerros Bravos and Mina Vieja was isolated above 580°C, suggesting it is carried by hematite. Therefore, to ensure that we are comparing similar age magnetizations across the formation, the hematite component of the Quebrada Ciénaga rocks is preferred as the ChRM for the remainder of the discussion.

At all three localities, site mean directions group better after tilt correction, although the improvement is insufficient to pass a fold test at the 95% confidence level [*McElhinny*, 1964]. This lack of clear positive fold tests is a consequence

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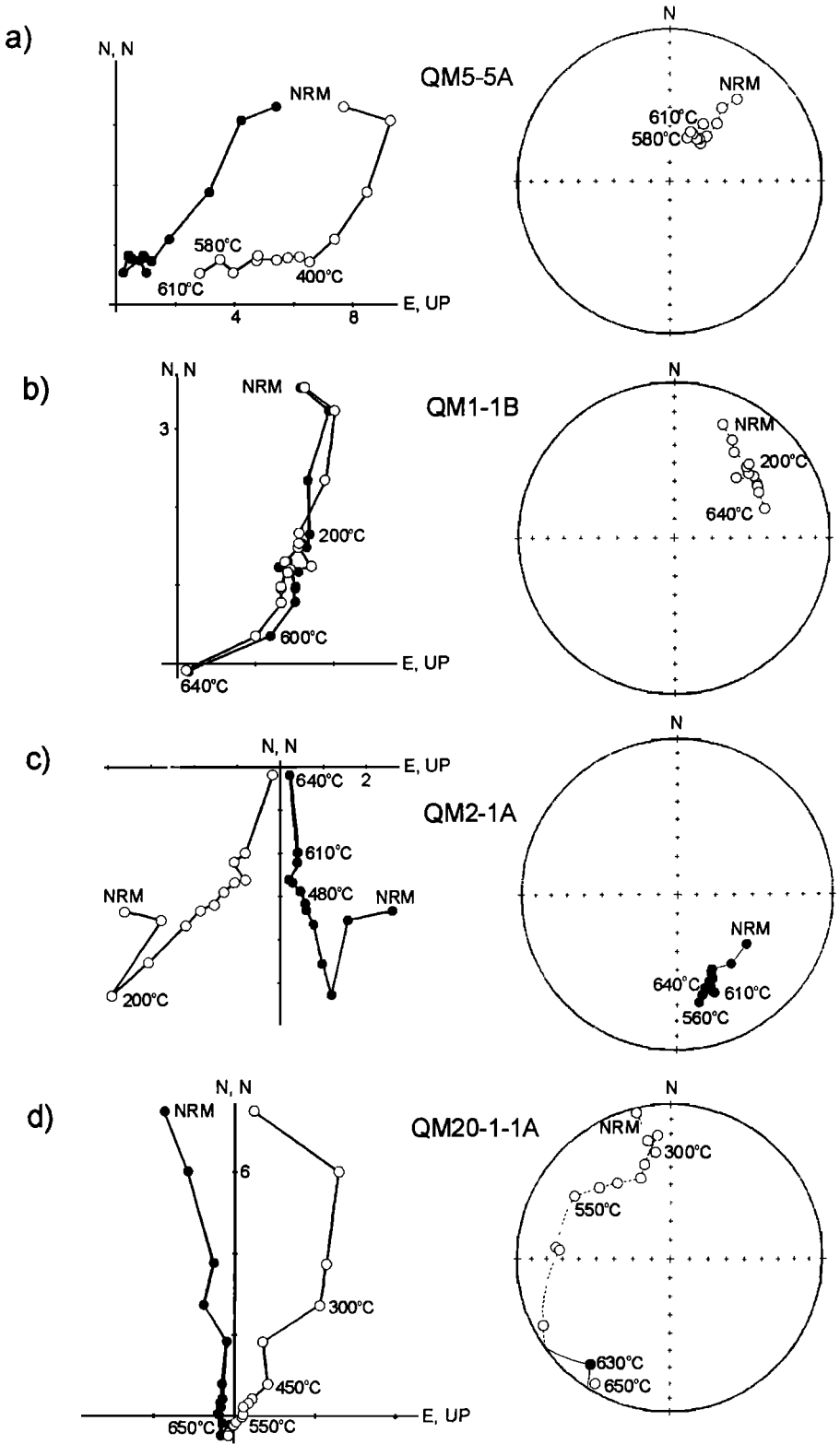


Figure 7. Representative orthogonal and equal-area projections for samples from the Quebrada Monardes Formation. In orthogonal plots, open (solid) circles represent vertical (horizontal) projections; intensities are in mAm^{-1}. On stereoplots, open (solid) circles represent negative (positive) inclinations. All plots are shown tilt corrected.

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Table 2. Results From the Quebrada Monardes Formation[a]

		Uncorrected			Tilt Corrected			
Site	n/N	Dec.	Inc.	TC	Dec.	Inc.	k	α95
				Quberada Cerros Bravos Region				
QM1	5/7	52.5	-34.5	090/10	47.8	-27.9	32.9	13.9
QM2	5/6	174.6	60.2	085/8	174.7	52 2	58 7	10 1
QM3	6/6	25.0	-35.5	121/12	25 7	-23.6	19.6	17.2
QM4	6/7	26 6	-35 1	126/10	27.5	-25.2	40.6	11 1
QM5	6/7	3 1	-60 4	155/12	18.1	-53.3	36.9	12.0
QM6	4/6	29 6	-52 7	173/15	41.8	-42.4	38 0	15.1
QM7	4/5	6 6	-27 9	230/26	22 0	-43 4	29 0	17.3
QM8	7/7	194 9	41 4	202/20	212 5	40 8	17 8	14 9
Mean IS	8/8	20.8	-44.7				20 4	12.5
Mean TC					27 8	-39.5	23.8	11 6
				Mina Vieja Region				
QM9	6/6	305.3	4.8	181/34	308 6	-23 0	34.1	12.1
QM10	6/6	309.9	-11.2	185/37	321.9	-39.7	14.0	18.9
QM11	7/7	315.7	-37.9	160/31	343 1	-44 0	44.4	9.2
QM12	4/5	325 4	-24.7	160/30	340 7	-28 5	22.6	22 8
QM13	6/6	304.0	-28.7	159/37	330.1	-43.3	45 7	10.0
QM14	3/4	309.7	-13.6	182/28	318.3	-34.7	79 1	14.0
Mean IS	6/6	311.5	-18.7				23.5	14 1
Mean TC					326 6	-36.2	34.2	11 6

[a] Notation is as for Table 1.

of the shallow dips and relatively consistent strike azimuths at each locality. Statistical comparison of the formation mean directions from the three localities does, however, pass an inclination-only fold test. In this test the fact that the angle between the paleomagnetic remanence direction and the bedding plane must remain constant during any rigid-body rotation can be used as a tilt test on a regional scale, even where the localities have different rotation histories [*McFadden and Reid*, 1982]. Thus comparison of in situ and tilt-corrected paleomagnetic inclinations can be made from our three different areas that appear to have different rotation histories as indicated by their declinations. The effect of progressive untilting on the precision parameter k of the three formation mean inclinations, following the statistical method of McFadden and Reid [1982], is shown in Figure 9. The increase in k after correction exceeds the 99% confidence level for an F test distribution and thus constitutes a statistically significant improvement and hence a positive test. We therefore interpret this magnetization as a prefolding remanance and not a postfolding remagnetization.

5.3 Jurassic Formations

The majority of samples from the Jurassic formations failed to yield stable directions or geologically interpretable primary remanence directions (Table 4). For the most part, samples showed two types of behavior: in situ northerly directions with moderate upward inclinations, consistent with a viscous remanence acquired in the present field direction; or directions that are inconsistent with either the present field direction or the ChRM identified from other units when considered either in in situ or tilt-corrected coordinates. Samples without obvious present-day components typically had high total NRM intensities that decayed rapidly when subjected to AF demagnetization, reminiscent of the effects of lightning strike-induced isothermal remanent magnetizations. Tilt-corrected site mean directions in some of the Quebrada Vicuñita sequence sites have a tendency toward NE directed declinations with moderate upward inclinations or, in one case (site QV7), its antipode, consistent with some of the data recorded in the younger units. The in situ directions of these sites, however, fall close to the Earth's present field direction, and the site mean directions overall have increased scattering on tilt correction, undermining our confidence in the directions having any geological significance. Only four sites from the Sierra Fraga Formation group sufficiently well after tilt correction to pass a fold test [*McElhinny*, 1964]. The confidence limits on both the individual site mean directions and the resultant formation mean coupled with the shallow inclination of the tilt-corrected formation mean, which implies major north-south displacement, make us skeptical of the significance of these data.

5.4 Identification of Magnetic Carriers

The nature of the magnetic carriers in all units was investigated using acquisition of isothermal remanent magnetization (IRM) and subsequent thermal demagnetization of three component IRM with along-axis fields of 800, 300, and 50 mT [*Lowrie*, 1990]. Additionally, high-temperature susceptibility behavior was also examined for some samples.

IRM experiments on the Cerro Valiente sequence demonstrate behavior consistent with a mixture of low- and high-coercivity carriers in varying proportions. This is consistent with the demagnetization behavior of the NRM and suggests that magnetite and hematite are carriers of natural

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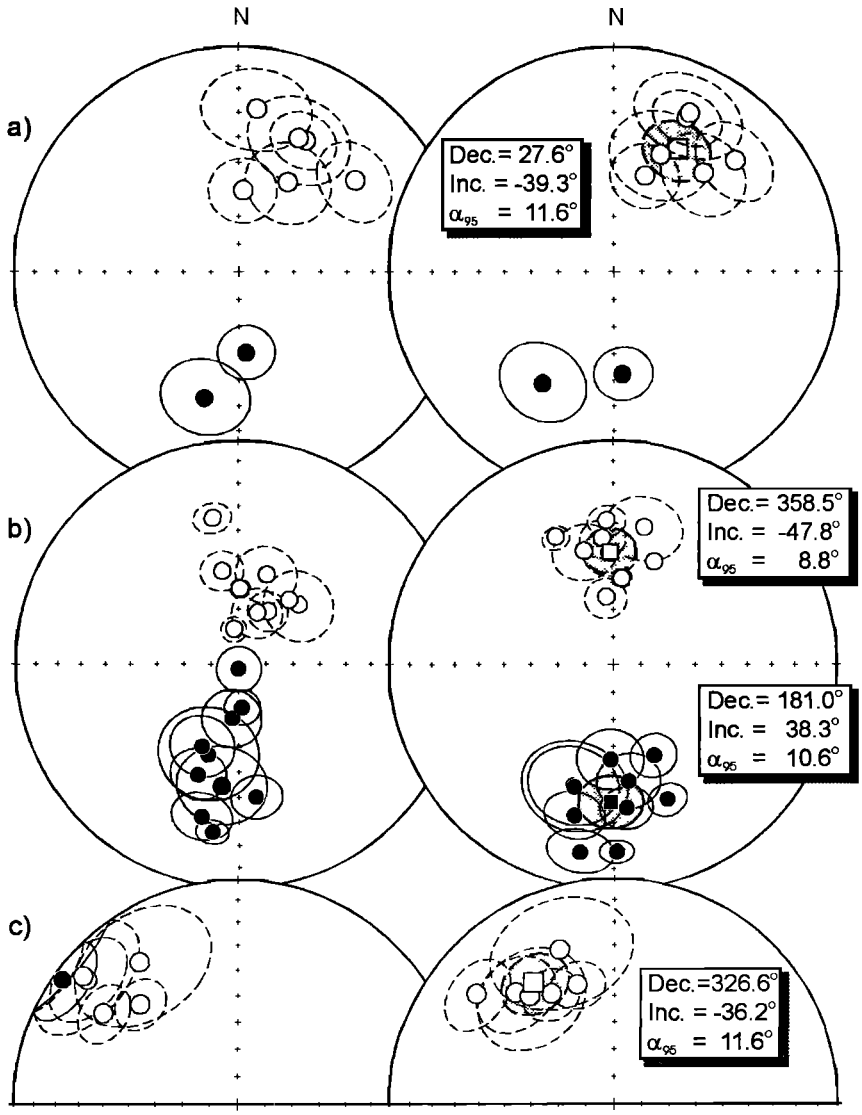


Figure 8. Equal-area projections summarizing the paleomagnetic data from the Quebrada Monardes Formation. The data are shown in (left) in situ (geographic) coordinates and (right) tilt-corrected coordinates. (a) Quebrada Cerros Bravos region, (b) Quebrada Ciénaga (with both normal and reverse polarity components shown), and (c) Mina Vieja region. Open (solid) circles represent negative (positive) inclinations with their associated 95% confidence limits. The formation mean direction is represented by a square, and its 95% confidence limit is shaded. Triangles denote sites excluded from the formation mean.

remanence as well as induced magnetizations. As most of the drilled samples broke up explosively on heating, high-temperature susceptibility studies of powdered samples were more successful in confirming the high-coercivity carrier as hematite.

Visual inspection of samples from the Quebrada Monardes Formation shows a range of red coloration; IRM experiments were therefore performed on a broad range of samples, and end-member examples are shown in Figure 10. The samples display a mixture of low- and high-coercivity carriers in varying proportions that demagnetize at ~580° and 670°C, respectively, consistent with a mixture of magnetite and hematite carrying IRM and NRM.

6. Discussion and Interpretation

6.1 Interpretation of the Mean Directions

In the Cerro Valiente sequence, seven of 10 sites in the central and all nine sites from the southern domain carry a ChRM. The demagnetization behavior from this unit is relatively simple, and the within-site variation is much less than that observed in the other formations. Site directions from the central domain pass a fold test, indicating a prefolding remanence. Uniformity of the structural corrections in the southern domain prohibits a meaningful fold test. The inclination of the formation mean direction from the south is, however, much closer to that from the

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Table 3. Results From the Quebrada Monardes Formation in Quebrada Ciénaga

	Low Unblocking Temperature Component							High Unblocking Temperature Component						
		Uncorrected		Tilt Corrected					Uncorrected		Tilt Corrected			
Site	n/N	Dec.	Inc.	Dec.	Inc.	k	α_{95}	TC	n/N	Dec.	Inc.	Dec.	Inc.	k	α_{95}
QM15	7/7	38.8	-59.9	21.9	-49.1	648.8	2.4	072/16	6/7	193.7	30.3	190.2	16.4	47.4	9.8
QM16	6/6	20.7	-70.5	358.7	-48.5	55.5	9.2	069/25	6/6	187.0	69.7	172.6	46.1	65 8	10.8
QM17								081/16	6/6	200.1	46.1	194.4	31.0	56 1	9.7
QM18	7/7	1.5	-62.5	6.1	-58.2	306.8	3.5	125/5	7/7	172.3	39.4	174.8	65.7	57.5	8 6
QM19	6/7	349.9	-34.1	357.6	-36.0	117.3	6.2	184/11	7/7	187.9	43.9	198.1	42 1	17.8	15.3
QM20	6/6	350.1	-55.3	355.6	-43.1	78 5	7.6	104/13	6/6	198.6	54.7	197.6	41.8	20.8	17.7
QM21	6/6	29.7	-68.1	354.0	-65.9	83.9	7.4	020/15	5/6	203.9	56.7	181.7	54.8	69 1	11.5
QM22	7/7	46.5	-59.2	12.9	-37.6	27.3	12.9	065/35	7/7	174.6	88.0	156.2	53.1	55.7	8.2
QM23	6/6	352.4	-77.5	336.0	-38.3	246.1	4.6	060/40	6/6	175.0	74.0	158.2	35.1	106.3	6 5
QM24	6/6	17.8	-5.6	346.0	-47.1	38.9	10.9	020/26	6/6	189.0	24.9	179.1	17.6	170.9	5.1
Mean IS	9/10	12.8	-62.0			25.0	10.5		10/10	190.2	53.1			15.4	12.7
Mean TC				358.5	-47.8	35.0	8.8					181.0	38.3	21.7	10.6

[a] Notation is as for Table 1

central region when both directions are considered in tilt-corrected coordinates. This, combined with the presence of sites with opposing magnetic polarities, suggests that the Cerro Valiente rocks in the southern domain also carry a prefolding remanence and not a postfolding remagnetization.

The reason for the failure of three sites in the central domain to record the ChRM is uncertain. Some samples carried very high NRM intensities (up to 2400 mAm^{-1}) and had low median destructive fields (typically, below 15 mT), suggestive of possibly having been remagnetized by lightning strikes. It is noteworthy, however, that the three site mean directions group better before application of a tilt correction (Figure 6) and therefore may be recording a later, postfolding remagnetization. The origin of this remagnetization may be related to hydrothermal activity associated with emplacement of the Eocene porphyry copper-related intrusions. If the remanence recovered is a postfolding remagnetization, the consistency of its directions with the prefolding remanence recovered from the other sites (Figure 6) suggests that both were acquired prior to any tectonic rotation of the central domain.

The demagnetization behavior of samples from the Quebrada Monardes Formation is variable and complex, but a ChRM was recovered from sites at all three sampling localities. The positive result from the inclination-only tilt test coupled with normal and reversed polarity site mean directions at two of the three sampling localities provides strong evidence that the remanence direction recovered from the Quebrada Monardes Formation is prefolding in origin. The presence of both polarity components in nine of the 10 sites from Quebrada Ciénaga is tentatively interpreted as being a combination of a detrital magnetization carried by magnetite and an early diagenetic magnetization carried by hematite.

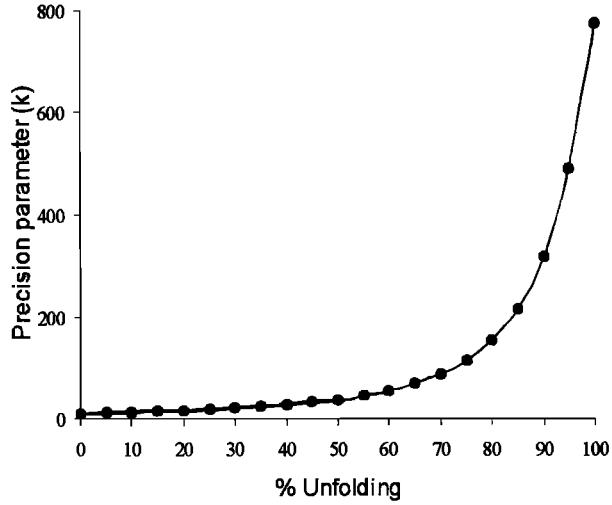


Figure 9. Plot of precision parameter *k* against percent unfolding for the formation mean inclinations of the Quebrada Monardes Formation. The plot demonstrates the improved clustering of the formation mean inclinations after 100% untilting, following the method of *McFadden and Reid* [1982].

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Table 4. Results From Sites Excluded From Further Discussion[a]

		Uncorrected			Tilt Corrected			
Site	n/N	Dec	Inc.	TC	Dec.	Inc	k	α_{95}
Quebrada Vicuñita sequence - Central Domain								
QV1b	3/6	185.3	51.5	031/64	155.3	5 8	12 9	19 4
QV2b	6/6	355.0	-35.7	031/64	342 8	10.0	31.1	12.2
QV3b	3/6	356.8	-49.6	325/81	273.2	-27.2	230.3	8.1
QV4	4/6	16.5	-60.3	151/22	33 0	-42.3	15.7	17.5
QV5	6/6	7.0	-44.7	151/22	19.5	-29.7	15.2	17 7
QV6b	6/6	271.2	-53.5	232/73	177.0	-36.4	13.4	19 0
QV7	6/6	167.2	40.6	186/40	203.8	41 0	100.3	6.5
QV8	6/6	358.2	-50.0	205/28	35 2	-54.3	48 9	9.7
QV9	6/6	348.3	-47.8	205/28	24.3	-57.4	175.6	5.1
QV10	3/6	9.3	-44.3	197/23	31.6	-42.9	61.8	15 8
Mean IS	6/10	0.3	-48.4				61.6	8.6
Mean TC					27.5	-44.7	54.8	9.1
Quebrada Vicuñita sequence - Southern Domain								
QV11	5/6	347.5	-61.4	165/16	13.7	-57.0	145.8	7.2
QV12b	6/6	69.5	62.1	165/16	62.6	77.9	49.0	10.4
QV13b	6/6	327.4	-48.2	075/25	332.3	-24.0	474.3	3.1
QV14	6/6	1.1	-49.9	169/9	10.9	-47.3	177.9	5 0
QV15b	6/6	347.8	-24.4	131/16	352.2	-14.3	277.1	4.0
QV16	6/6	354.8	-54.3	147/20	12.9	-42.1	46.3	10.0
Mean IS	3/6	355.1	-55.3				136.4	10.6
Mean TC					12.4	-48.8	113.3	11.6
Sierra Fraga Formation - West of Agua Amarga Thrust								
SF1	6/6	137.1	47.3	005/64	122.3	-7.5	60.9	8.6
SF2b	6/6	75.6	55 6	005/62	84.1	-4.8	12.0	20.9
SF3	5/6	299.1	-24.8	005/62	301.0	32.3	16.0	20.3
SF4	5/6	104.5	79.2	190/56	278.8	44.8	13.2	24.6
SF5	3/6	314.0	51.5	192/54	301.3	1.9	62.7	15.7
SF6b	6/6	234.4	4.6	182/46	226.9	-30.8	22.7	14.4
SF7b	7/7	53.9	-10.1	181/46	49.5	26.3	10 9	19.3
Mean IS	4/7	311.0	16.8				1.6	122.7
Mean TC					297.1	21.8	13.1	26.3

[a] Notation is as for Table 1.
[b] Indicates sites excluded from the site mean calculation.

Some sites in the Quebrada Vicuñita sequence and Sierra Fraga Formation show the influence of lightning strikes. The reason for the failure of other sites to yield reliable paleomagnetic directions in these Jurassic units is less certain. Thin section analysis did, however, reveal pervasive alteration of these rocks with a mineral assemblage consistent with the effects of hydrothermal fluids.

6.2 Comparison with Paleomagnetic Reference Directions

The apparent polar wander path for South America remains the subject of much debate owing to the poor constraints on many of the available poles [e.g., *Beck*, 1988, 1998; *Randall*, 1998]. As the remanences recovered in this study predate the Eocene folding and are regarded as primary, the Cerro Valiente sequence and Quebrada Monardes Formation mean directions need to be compared with Paleocene and Early Cretaceous poles, respectively.

Comparison of the formation mean directions with directions calculated from appropriate reference poles [*Randall*, 1998] clearly demonstrates significant and variable amounts of rotation with, in most cases, insignificant latitudinal displacement (Table 5). Both formation mean directions from the Cerro Valiente sequence indicate statistically significant southward latitudinal transport, contrary to the insignificant shift suggested by the Quebrada Monardes Formation data and the regional geologic and paleomagnetic data sets, which indicate margin-parallel transport in the central Andes is negligible [e.g., *Beck* et al., 1994; *Randall*, 1998]. This apparent inconsistency with the Paleocene data may result from a problem with the Paleogene reference pole determination [*Randall*, 1998]. When compared with other proposed reference poles for the Paleogene or Paleocene [e.g., *Beck*, 1988; *Butler et al.*, 1991; *Roperch and Carlier*, 1992], the amount of latitudinal displacement is variable and sometimes statistically significant, suggesting that the available Paleocene reference poles may not be well determined yet.

The results from both localities in the Cerro Valiente sequence reveal statistically significant clockwise rotations. Two of the three studies in the Quebrada Monardes Formation

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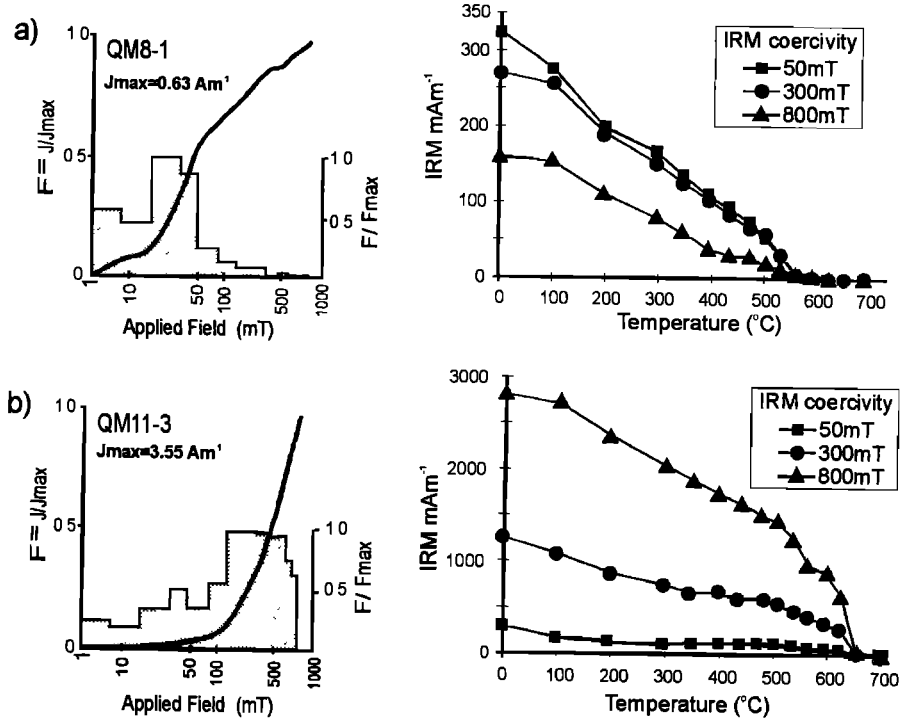


Figure 10. Isothermal Remanent Magnetization (IRM) acquisition and demagnetization curves for samples from the Quebrada Monardes Formation, following the technique of *Lowrie* [1990]. Applied fields are up to 800 mT and are plotted on a log scale. Both samples indicate a mixture of magnetite and hematite carriers with (a) magnetite dominant and (b) hematite dominant.

likewise demonstrate statistically significant rotations: clockwise in the Quebrada Cerros Bravos and counterclockwise in the Mina Vieja localities. The result from Quebrada Ciénaga indicates no statistically significant rotation. Irrespective of the reference pole used, the sampling localities in the Cerro Valiente sequence record clockwise rotations of larger magnitude than the Cretaceous rocks.

7. Tectonic Implications

The variation in the magnitudes and senses of rotation observed in this study cannot be explained by a single large-scale mechanism, such as oroclinal bending or domino-style block rotation by widely separated transandean faults, and instead indicate that local tectonics dominate the pattern of rotations. The greater clockwise rotations in the Cerro Valiente sequence could be interpreted to indicate a multistage history of rotations with an earlier counterclockwise rotation of the Quebrada Monardes Formation. However, the spatial consistency of the rotations with the regional structures suggests the pattern can be best explained in terms of a single post-Paleocene tectonic event where the rotation is localized and due to deformation related to the SC-AA Fault System and its subsidiary fault domains. It is impossible to discount a contribution from a large-scale process which would add a uniform rotation to the different

Table 5. Calculation of Rotation and Flattening[a]

Locality	Mean Direction Dec.	Inc.	α_{95}	Rotation	Flattening
	Quebrada Monardes Formation				
Quebrada Cerros Bravos	27.8	-39.5	11.6	31.3±12.6	-0.3±10.3
Mina Vieja	326.6	-36.2	11.6	-29 9±12.1	-3 6±10.3
Quebrada Ciénaga	1.0	-38 3	10 6	4.5±11.4	-1.5±9.6
	Cerro Valiente Sequence				
Central domain	40.1	-38.5	12 1	35.2±17.2	-16.0±13.6
Southern domain	47.5	-40 5	10.7	42.6±16 4	-14 5±12 8

[a] Rotation and flattening are given in degrees using the parameters of *Beck* [1980] and *Demarest* [1983]. Positive rotations are clockwise, and positive flattenings indicate northward latitudinal transport. Reference directions are calculated from the poles given by *Randall* [1998] using a location of 69.4°W, 26 5°S, close the center of the studied location. The reference directions lie at Dec. = 356.5°, Inc. = -39.8°, and α_{95} = 4° for the Quebrada Monardes Formation and Dec. = 355.9°, Inc. = -55.3°, and α_{95} = 12° for the Cerro Valiente sequence.

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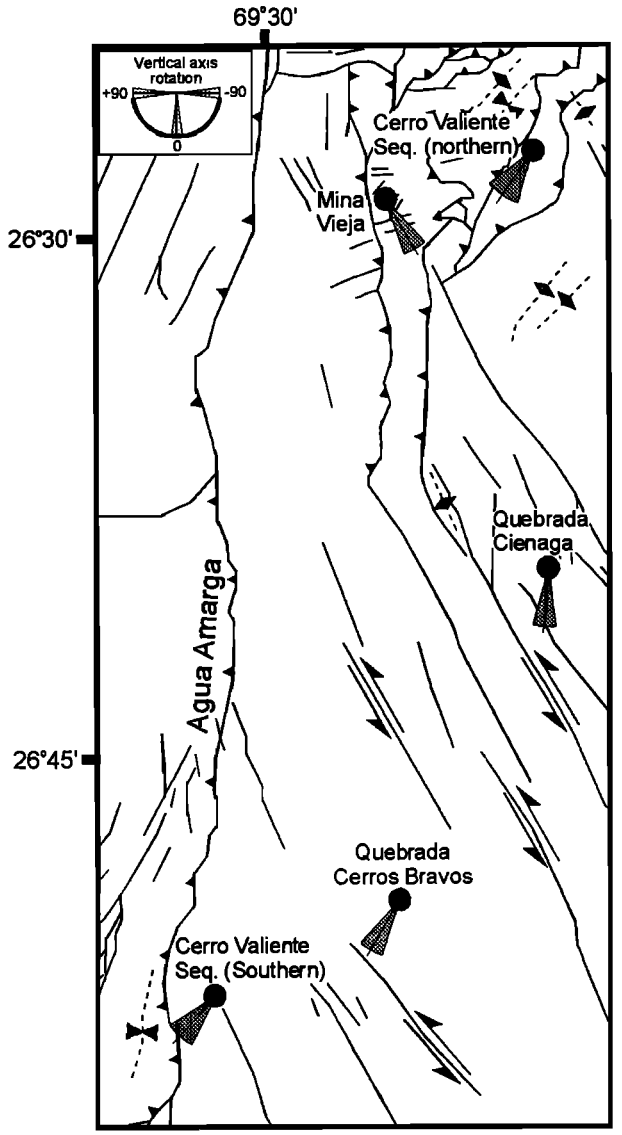


Figure 11. Distribution of rotations and major faults in the study area.

localities, but the variation in the amounts of rotation can only be explained by local controls. Furthermore, the magnitudes of the clockwise rotations greatly exceed those predicted by current oroclinal bending models. The driving force for the rotations was apparently the major regional sinistral transpressive phase associated with deformation of the arc at ~42-32 Ma [*Tomlinson et al.*, 1993; *Mpodozis et al.*, 1994; *Cornejo et al.*, 1997].

Within the southern domain the amount of rotation recorded by the Cerro Valiente, Quebrada Cerros Bravos, and Quebrada Ciénaga sampling localities increases toward the SC-AA Fault (Figure 11). This implies a gradient in the deformation across the southern domain, at least with respect to vertical axis rotations, wherein deformation increases toward the master fault. Similar deformation and paleomagnetic rotation gradients have been observed in the borderlands of other transcurrent fault systems and have been modeled analytically and *numerically [Nelson and Jones*, 1987; *Sonder et al.*, 1986]. Following such models, the rotation gradient in the southern domain indicates the rotations are related to activity of the SC-AA system, but furthermore, given the sinistral sense of the simple shearing component of the master fault, the clockwise sense of the rotations suggests the rotations are a result of the pure-shear shortening component of the deformation. Although the above cited numerical models treat only the classic (nontranspressive) wrench-type fault systems, in theory the same gradients can occur for the shortening component in transpressive systems [*Dewey et al.*, 1998].

The lack of a paleomagnetically determined rotation in the Quebrada Ciénaga area, despite being located where the NW trending strike-slip faults have the greatest displacements, suggests that these faults did not produce significant rotation. The lack of rotation may be related to a number of factors, including the faults are widely spaced (2-4 km), their slips are small (hundreds of metres to 1-2 km), and they appear to act as a transfer zone between two belts of shortening, suggesting the shear zone boundaries may be parallel to the faults themselves, in which case there would be no domino-style rotation associated with them. The NW trending faults maintain a more or less constant strike across the southern domain, despite the variation in paleomagnetic rotations, suggesting those faults in the Quebrada Cerros Bravos and Cerro Valiente localities developed late in the deformation history.

One alternative hypothesis is that the southern domain suffered a more or less homogeneous clockwise rotation and that sinistral shear on the NW trending faults in the Quebrada Ciénaga area overprinted a counterclockwise rotation. The hypothesis cannot be excluded, but it seems less probable in that it requires the fortuity that the overprinting counterclockwise rotation be of similar magnitude as the clockwise rotation.

Abels and Bischoff [1999] recently proposed a model wherein widely spaced NW trending basement faults transect the entire Eocene forearc and produce clockwise declination anomalies by a domino-style rotation of large crustal blocks. As their model is largely based on the geology of the southern domain, the data contained herein pertain directly to their model. In their model, one of their NW trending transandean faults corresponds to the set of NW trending faults in the Quebrada Ciénaga area. The mapping of *Cornejo et al.* [1993, 1998] indicates, however, that the faults in Quebrada Ciénaga do not cut the SC-AA Fault, nor do they cut stratified units located in the hanging walls of the reverse faults which the NW trending faults terminate into; instead the stratified units strike across the ends of the NW trending faults (Figure 2). We do not dispute *Abels and Bischoff*'s [1999] suggestion that the Eocene NW trending faults may have had an origin in the reactivation of older basement structures that have a greater continuity, but we do dispute that these basement faults were reactivated across the entire width of the forearc as Eocene transandean faults. Instead, the mapping indicates that the Eocene activity of the NW trending faults is localized and that they are secondary to the main N-S trending structures in the area. Furthermore, although the basic form of the *Abels and Bischoff* [1999] model suggests uniform

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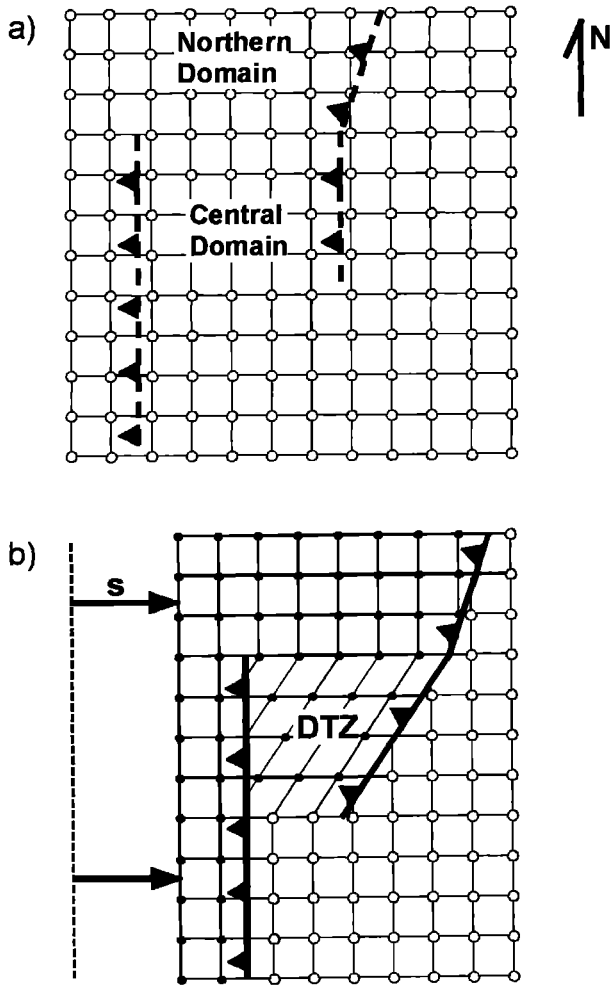


Figure 12. Kinematic model for clockwise rotations in the central domain. (a) Nondeformed reference grid (plan view) with inferred initial orientation of reverse faults in the northern and central domains. Since there is no paleomagnetic data from the northern domain, the faults there are shown, for simplicity, as having initiated with their current strike (18°) and maintained that orientation throughout the deformation. In the central domain the initial structural orientation is shown as N-S, consistent with their approximate orientation if the currently NE trending structures are rotated back to undo the declination discordance. Note the locus of reverse faulting changes across the central domain from a position in the northeast to a position in the southwest. (b) Deformed grid after shortening *s*, which is constant along the length of the grid. At the latitude of overlap of the two thrusts, displacement on the lower eastern thrust decreases to the south while displacement on the higher western thrust increases from zero displacement at its northern tip, producing a displacement transfer zone (DTZ). As a result of the differential movement of the lower thrust, the fault and hanging wall units rotate clockwise, and E-W dextral shear is generated.

clockwise rotations, they point out that the regional pattern of paleomagnetic data is more variable. They suggest that the variability in paleomagnetic declinations can be explained if the rotating blocks are internally deformed and fragmented into smaller size blocks. However, in order for their model to describe the paleomagnetic pattern in the southern domain it requires, as in the second hypothesis above, the fortuitous situation that the margins of the rotating blocks deform sufficiently to yield no declination anomaly (Quebrada Ciénaga). Instead, the hypothesis favored here is that the master Agua Amarga thrust is the source of maximum rotation-related deformation.

In the central domain the clockwise rotation obtained from the Cerro Valiente sequence is interpreted as the result of the kinematics of the displacement transfer that characterizes the area (Figure 12). In the kinematic model, differential movement on lower thrusts, related to the progressive replacement of their displacement by movement on higher, westerly positioned thrusts, causes clockwise rigid-body rotations and E-W dextral shear manifested by numerous E-W dextral strike-slip faults. The model is consistent with the notably more NE trend of structures in the domain compared to those in the northern fold-and-thrust belt and in the southwestern sector of the central domain (Figure 3). In fact, if the NE trending structures are rotated back to undo the declination anomaly indicated by the paleomagnetic data, the structures become N-S trending, parallel to the structural grain in the southwest sector (Figure 12a).

The counterclockwise rotation at the Mina Vieja sampling locality is consistent with the structural trends in the immediate area (Figures 3 and 4) and the prediction of *Cornejo et al.* [1993]. If the structural trends are restored by the rotation amount (~30° clockwise), the map-scale fault bend anticline becomes N-S trending, and the dextral strike-slip faults become E-W striking, parallel to the correlative structures in the surrounding area. This suggests the N-S structural grain in the surrounding area is a primary orientation with no significant paleomagnetic rotation.

Crosscutting relations indicate that the counterclockwise rotation of the Mina Vieja region occurred during the Eocene deformation. Since the rotation affected the trend of the fault bend anticline and dextral strike-slip faults, it must postdate the generation of these Eocene structures. Late in the deformation history, the ~36 Ma Potrerillos Cobre porphyry intrusion [*Marsh et al.*, 1997] was emplaced in the core of the anticline and contains in its contact aureole a metamorphic foliation having an orientation consistent with the regional shortening direction [*Tomlinson*, 1994]. The orientation of the cleavage is the same both east of the intrusion, where bedding strikes N-S, and northwest of the intrusion, where the cleavage transects the counterclockwise-rotated fault bend anticline (Figure 4). Maintenance of the cleavage orientation across the area suggests the counterclockwise rotation precedes the cleavage formation at ~36 Ma.

The mechanism by which the counterclockwise rotation was accomplished is less clear as it is inconsistent with the kinematics of the central domain described above. Furthermore, since the rotation is highly localized it is unlikely to be a result of the SC-AA system-parallel sinistral simple shearing. *Cornejo et al.* [1993], in addition to describing the E-W trending dextral faults, reported several E-W trending sinistral faults. They interpreted the sinistral movements to represent reactivation of the dextral faults during a second, probably unrelated, event. We propose the

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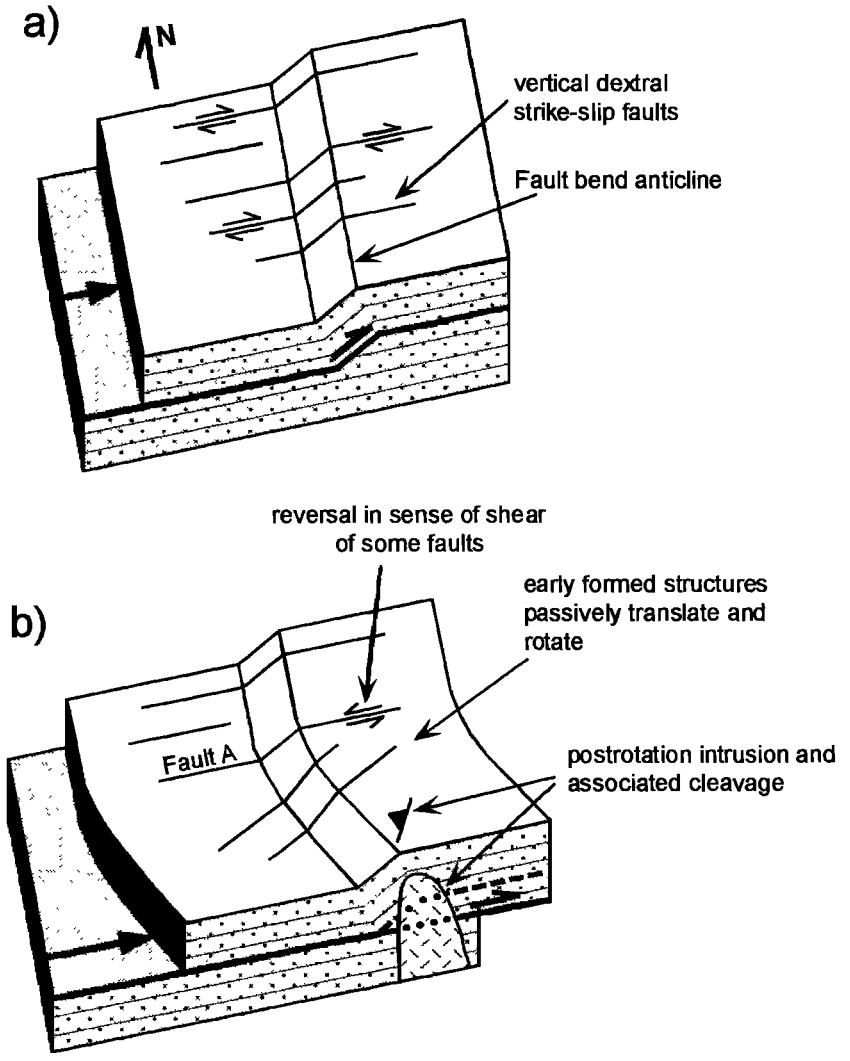


Figure 13. Block diagram modeling formation of the counterclockwise rotation at the Mina Vieja locality. (a) Inferred prerotation orientation of the structures in the area. (b) Impedance of thrust displacement on the north but down stepping and continued displacement of the thrust on the south results in rotation of the hanging wall fault bend anticline and associated minor faults. After rotation, the 36 Ma Potrerillos Cobre porphyry intrusion is emplaced into the core of the anticline, and a cleavage forms in its contact aureole, crosscutting the rotated anticline.

sinistral movements were produced during the same progressive Eocene deformation and reflect, at least locally, a reversal in the kinematics of the central domain. Figure 13 illustrates how this might have been accomplished. The fault bend anticline formed initially as a N-S trending structure above a ramp flat transition in the underlying thrust (Figure 13a). At some point in time, the southern part of the thrust may have stepped down into the footwall block and began translating the anticline and its underlying ramp passively eastward, while the northern part of the thrust maintained its higher position and was impeded from further displacement (Figure 13b). The different levels of the thrust would be connected by a lateral ramp having E-W sinistral shear, above which the fault bend anticline and geologic units would rotate counterclockwise. In contrast, the Bailey Willis reverse fault, located to the west, maintains its N-S strike (Figure 4), suggesting that either it had reciprocal and compensatory differential slip contemporaneous with the sinistral shear in the footwall or postdates the footwall rotation as an out-of-sequence reverse fault. Exploration drilling [*Reyes*, 1981] and structural restorations [*Tomlinson*, 1994] suggest the fault bend anticline formed above a frontal ramp lying over a step in the Paleozoic igneous basement. Such irregularities in the underlying basement or other mechanical heterogeneities in the footwall block may have prevented or hindered down stepping and displacement of the thrust on the north side, leading to development of the E-W sinistral shear and counterclockwise rotations. A similar model, but on a larger scale, has been proposed for the Miocene clockwise rotation of regional structures in the subandean fold-and-thrust belt in southern Bolivia [*Kley*, 1996; *Somoza et al.*, 1999].

This is the first well-constrained counterclockwise rotation

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to be recorded in the southern central Andes. *Hartley et al.* [1992b] report data from the Mejillones Peninsula which likewise suggest counterclockwise rotations. However, many of the site mean directions are poorly determined statistically, and the range of rotations among sites is so large that they did not calculate a mean for the formation or area. They attributed the rotations to small-scale fault block rotations, with each site representing an independent record of rotation. The Mina Vieja rotation is apparently highly localized, and so no inferences can be drawn from this data about the large-scale tectonics of the studied area or the tectonic history of the Andean margin, but it does serve to demonstrate the importance of local tectonics for rotations in the south central Andes.

This is now one of several studies in the Andean margin where detailed paleomagnetic sampling over small structurally well-constrained areas has revealed complex patterns of rotation. Other examples from the Argentine Puna [*Aubry et al.*, 1996] and the Bolivian Altiplano [*Butler et al.*, 1995] have also shown temporal and spatial variability of rotations. Together these integrated studies highlight the importance of local tectonic influences on the observed rotation patterns and demonstrate that an understanding of local structure is crucial to the interpretation of rotation data. Furthermore, care should be taken when using paleomagnetic data to interpret the large-scale tectonic evolution of the Andes and other tectonically active margins.

8. Conclusions

The presence of dual magnetic polarities at four of the five sampling localities, a positive inclination-only fold test in the Quebrada Monardes Formation, and a positive fold test in the Cerro Valiente sequence indicates that all units carry a predeformational and probably primary remanence. Comparison with appropriate reference poles indicates varying amounts of rotation in the region east of the SC-AA Fault. The pattern is too complex to be explained by large-scale tectonic rotation mechanisms alone, whether it be oroclinal bending or large-scale domino-style block rotation. The model proposed to explain the interpreted rotations is therefore one of discrete, localized, in situ rotations resulting from an Eocene sinistral transpressive event wherein response to the transpressional deformation is varied and forms several distinct domains of differing tectonic style, rotation amounts, and rotation senses.

Results from the southern domain of sinistral strike-slip faults demonstrate a decrease in clockwise rotation with increasing eastward distance from the master fault trace, irrespective of the age of the sampled unit. This is consistent with the rotations being related to activity of the SC-AA Fault. Furthermore, the clockwise sense of the rotations is consistent with the structural style that indicates a dominance of shortening over margin-parallel shear in the transpressive system. Since units as young as Paleocene are affected, the rotations must be post-Paleocene in age. The result from the central domain likewise records a post-Paleocene clockwise rotation and is interpreted in terms of rotation due to differential motion on thrust sheets in a displacement transfer zone between the fold-and-thrust belt to the north and the N-S trending reverse faults, including the Agua Amarga thrust, to the south. Sites collected from the Quebrada Monardes Formation in the Mina Vieja area record a counterclockwise rotation that is highly localized and interpreted as a result of E-W sinistral shear developed during differential movement of the underlying thrust; the differential movement possibly being imposed by along-strike mechanical heterogeneities in the block through which the thrust propagated. The counterclockwise rotation affects Eocene structures but not a cross-cutting cleavage developed late in the deformation history and associated with the 36 Ma Potrerillos Cobre porphyry intrusion. These relations therefore bracket the timing of rotation to the Eocene.

Overall, the data highlight the importance of localized tectonics in controlling rotations, even where the rotations are the consequence of a single deformational event. Furthermore, it underscores the significance of relating rotations interpreted from paleomagnetic data in terms of the structure and tectonic history of the sampling locality.

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